Lab Report: Automatic Classification and Mixture Models

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In this lab, we will explore the `MetaHIT_SangerSamples.genus.txt` database used in studies on the gut microbiome, particularly those that led to the discovery of enterotypes. These enterotypes represent three distinct microbial ecosystems present in the human gut. We will then apply the unsupervised clustering techniques K-means and PCA, in order to attempt to reproduce these classifications. We will then visualize the results obtained, and evaluate the performance of these methods. These analyses will allow us to better understand the characteristics of enterotypes and their potential impact on human health.

1. Presentation of the `MetaHIT_SangerSamples.genus.txt` Database

This study is based on the database entitled `MetaHIT_SangerSamples.genus.txt`, which contains 249 rows, of which 248 correspond to different bacterial species, and one additional row, entitled "reads", listing other bacteria present in very small quantities. The database also contains 33 columns, each representing a studied individual. Although the exact content of the dataset is not specified, it most likely consists of measurements of the relative abundance of each bacterial species in each individual.

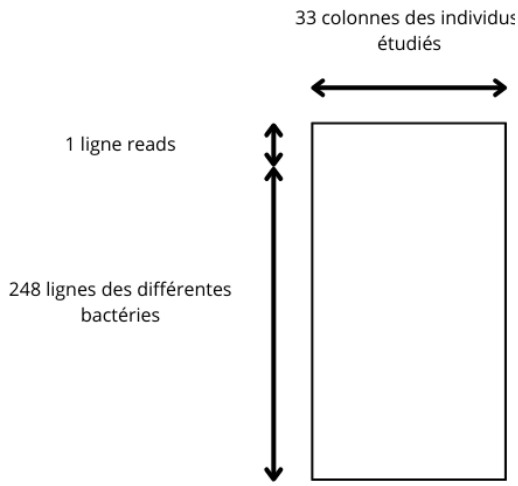


FIGURE 1 – Presentation of `MetaHIT_SangerSamples.genus.txt`.

As indicated in the problem statement, we will remove the row corresponding to "reads". Since our objective is to classify individuals, we will also transpose the data matrix before applying the different methods.

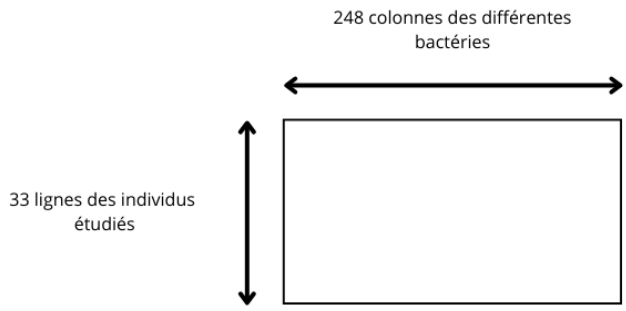


FIGURE 2 – Presentation of the transposed `MetaHIT_SangerSamples.genus.txt`.

2. K-means

2.1. Theoretical Explanation

Given a set of points $(\mathbf{x}_1, \ldots, \mathbf{x}_n)$, the objective is to partition these n points into K subsets $C = \{C_1, \ldots, C_K\}$ with $K \leq n$. K-means is a clustering method that groups observations into clusters, without using labels, based on their similarity. It is an exploratory unsupervised learning technique that allows data to be analyzed and interesting relationships between them to be highlighted.

2.1.1. Mathematical Foundations and Algorithm Operation

K is a hyperparameter defined in advance. The objective of K-means is to minimize the total inertia, defined as the sum of the intra-cluster variations $W(C_k)$ for all clusters $k = 1, \ldots, K$.

$$\min_C W \triangleq \min_C \sum_{k=1}^{K} W(C_k) \tag{1}$$

where

$$W(C_k) \triangleq \sum_{\mathbf{x} \in C_k} d(\mathbf{x}, \bar{\mathbf{x}}_k)^2 \tag{2}$$

$d(\cdot, \cdot)$ is generally the Euclidean distance. Each cluster C_k is represented by its centroid $\bar{\mathbf{x}}_k$, computed as the mean of the observations assigned to that cluster :

$$\bar{\mathbf{x}}_k \triangleq \frac{1}{|C_k|} \sum_{\mathbf{x} \in C_k} \mathbf{x} \tag{3}$$

The algorithm proceeds iteratively by assigning each point $\mathbf{x}$ to the cluster C_k whose centroid is closest, then recomputing the centroids until convergence or until a maximum number of iterations is reached. Since the initialization is random, we risk obtaining different results at each execution of the algorithm, which can lead us to local minima of the criterion we seek to optimize. To address this problem, we can run the algorithm several times and take the average of the results obtained.

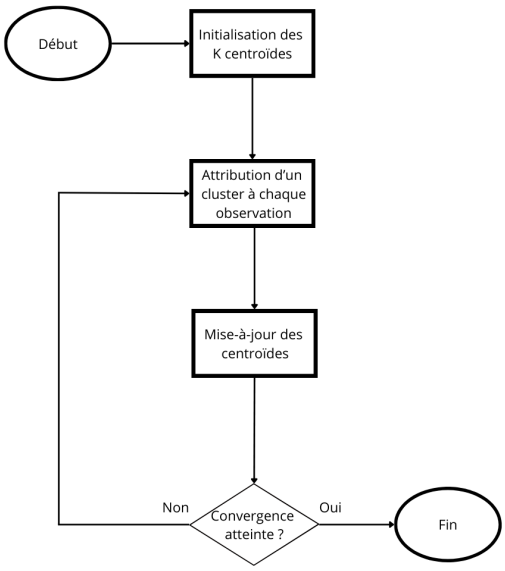


FIGURE 3 – K-means algorithm.

2.1.2. Performance Metrics

In order to determine the ideal number of clusters, we will use several metrics :

— **Inertia** : Inertia W measures the cost of using the method. We use the elbow method to determine the optimal number of clusters by analyzing the inertia curve as a function of the number of clusters. This method allows us to identify the trade-off between cost and number of clusters. Beyond a certain number of clusters, adding new ones does not lead to a significant improvement, either in the reduction of inertia or in the relevance of the model. Although each studied individual is unique, this approach does not allow recurring patterns in the data to be identified, which can lead to overfitting. The term "inertia" is used because it is homogeneous to a unit mass multiplied by a squared distance, and it is analogous to physical inertia.

— **Silhouette** : The silhouette S_K evaluates the quality of the segmentation of data into clusters by measuring the similarity of points within the same cluster relative to those of other clusters. Its value ranges from 0 to 1 : a value close to 1 indicates that the points of a cluster are homogeneous and distant from others, while a value close to 0 means that they are close to the boundary with other clusters. A high silhouette score is therefore desirable, as it indicates that the points are well matched to their own cluster and poorly matched to neighboring clusters.

$$S_K = \frac{1}{n} \sum_{k=1}^{K} |C_k| \times \overline{s_k}$$

with :

$$\overline{s_k} = \frac{1}{|C_k|} \sum_{i \in C_k} s(i) \quad \text{and} \quad s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

where :
— C_k is the k-th cluster.
— $a(i)$ is the average distance of a point i to the other points in the same cluster.
— $b(i)$ is the minimum of the average distances of point i to the other points in the other clusters.

To determine the optimal number of clusters, we can also use DBSCAN, a clustering method that does not require fixing the hyperparameter K. However, it should be noted that DBSCAN is based on entirely different principles. Another approach is to use dendrograms, which represent data in the form of a tree by connecting the closest points. The number of clusters can then be chosen by analyzing the structure of the tree. However, we will choose not to go further into these methods, as they would weigh down our study.

2.2. Results

2.2.1. Optimal Number of Clusters

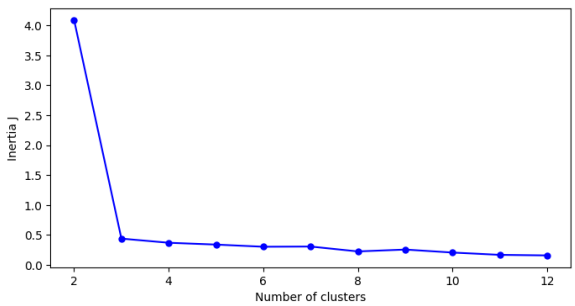


FIGURE 4 – Inertia as a function of the number of clusters.

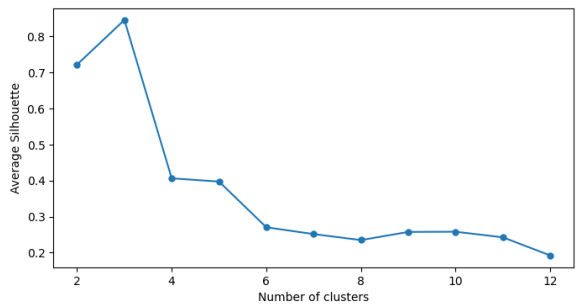


FIGURE 5 – Silhouette as a function of the number of clusters.

Both approaches, the elbow method applied to the inertia curve and the maximization of the silhouette, converge to indicate that the optimal number of clusters is 3. This confirms that 3 is the optimal choice for data segmentation.

2.2.2. Projection of Results

To visualize our results, we used a boxplot focusing on the most discriminating bacterium among the 248 studied. This bacterium was identified using an ANOVA test performed for each bacterium. *Prevotella* proved to be the most significant, with the lowest p-value of 8.95×10^{-16}, indicating a strong difference between the clusters. In an ANOVA test, the null hypothesis assumes that there is no significant difference between the clusters for a given bacterium. The lowest p-value shows the greatest statistical evidence against this hypothesis, meaning that the corresponding bacterium is the one that best distinguishes the clusters, thus facilitating the visualization of differences between groups.

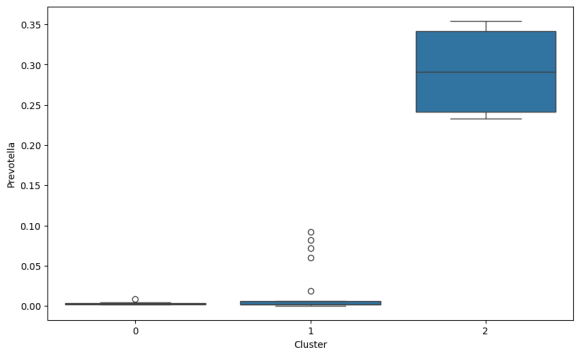


FIGURE 6 – Boxplot of the three clusters on the variable *Prevotella*, the most discriminating, determined by an ANOVA test on all bacteria.

The analysis of the boxplot for *Prevotella* revealed marked differences between the three clusters from the K-means analysis. Cluster 0 is characterized by a very low abundance of *Prevotella*, close to zero, with little variation. Cluster 1 also shows low abundance, similar to cluster 0, but with a slight increase in variability and some higher outliers. In contrast, cluster 2 stands out clearly with a significantly higher abundance, with a median around 0.29 and high variability, with values ranging from 0.23 to 0.35.

These results confirm that *Prevotella* is a strong marker of distinction between the clusters, particularly for cluster 2. The latter appears to represent a group of individuals with a gut microbiome characterized by an increased presence of *Prevotella*. The difference between clusters 0 and 1 is less pronounced, but the outliers in cluster 1 could indicate intermediate cases or individuals on the boundary between groups with low and high *Prevotella* abundance.

3. Raw Principal Component Analysis

To better understand the distinction between clusters 0 and 1, which appears less evident in the boxplot representation, we propose to perform a raw PCA without preprocessing of the database. This approach will potentially reveal other dimensions of variation that distinguish these two groups more clearly.

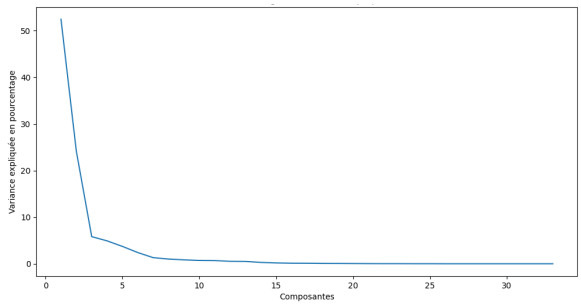


FIGURE 7 – Explained variance of the principal components for the raw PCA on a 33×248 data matrix. The first 33 components are retained because there are 33 individuals. The first component explains 52.46% of the variance, the second 24.12%, and the following ones have marginal importance. There is no size effect !

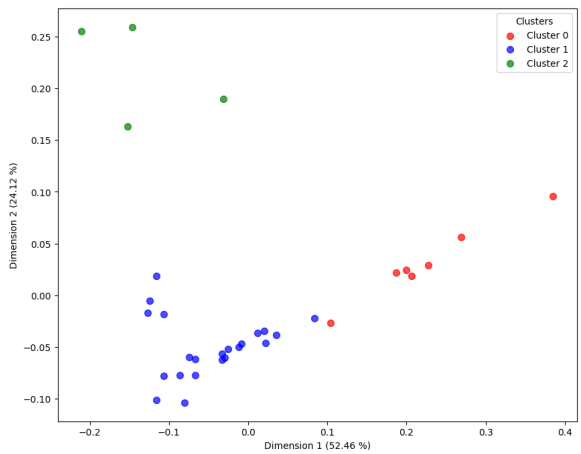
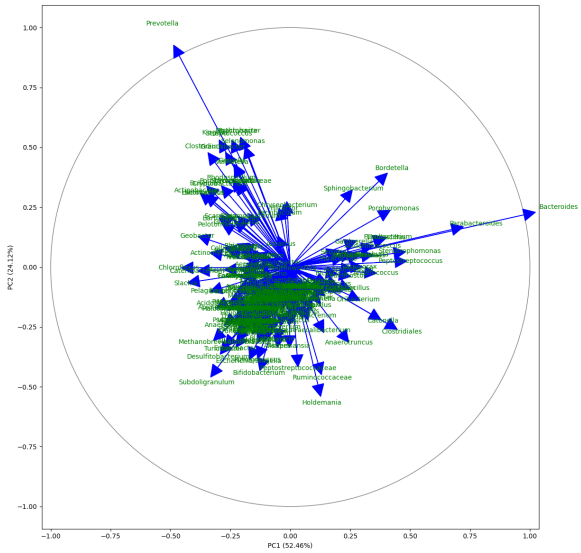


FIGURE 8 – Projection of data onto the plane defined by the first two principal components of the raw PCA.

The examination of the projections and the correlation circle reveals a clear structure in three distinct groups, confirming the relevance of our three-class clustering approach. The analysis of the correlation


Prevotella
Bordetella
Sphingobacterium
Pirellulomonas
Bacteroides
Parabacteroides
Peptococcus
Peptostreptococcus
Coprococcus
Anaerotruncus
Clostridiales
Holdemania
Ruminococcaceae
Bifidobacterium
Subdoligranulum
Methanobrevibacter
Desulfitobacterium
Geobacillus
Pelagibacterium
Actinomyces
Chlorobium
Slackia
Anaerostipes
Acidaminococcus
Pelomonas
Pelobacter
Kaistobacter
Clostridiales
Actinobacillus
Bacillus
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gut microbial ecosystems and open the way for future research on their impact on human health. This study also demonstrates the effectiveness of unsupervised clustering techniques and principal component analysis for exploring and characterizing complex biological datasets.